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# Synergistic integration between internet of things and augmented reality technologies for deaf persons in e-learning platform

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**Abstract**
The development of the Internet of Things (IoT) accentuates the interweaving of the digital with the physical, raising multiple issues, including for learning and more specifically for the learning of the deaf. While emerging technologies may have given rise to various methods and modes of learning, the implications of the IoT for deaf learning are not yet understood. This study aims to propose the possible integration of the IoT and augmented reality (AR) to support deaf learning from three dimensions of analysis: data, interfaces and pervasiveness. The axes of applications identified suggest that the IoT promotes learning characterized by experimentation, adaptation (to the context and to the learner), the manipulation of objects and exploration without the constraints of time or space. In order to achieve this integration, preliminary survey is collected from deaf learners. Based on the results, a deaf intelligent learning model is proposed and discussed.

**Keywords** Augmented reality · Communication · Deaf learners · E-learning · Emerging technology · Internet of things

## 1 Introduction

The availability of online learning resources has enabled deaf and hard of hearing (HH) students to access lectures [1, 2]. E-learning is a revolutionary way to prevent disability discrimination, as previously most disabled people in the community were unable to follow their education. However, for several reasons, online learning is not the preferred option for many deaf and HH people. Indeed, a person who grew up having sign language as the main means of communication can have difficulty understanding the concepts described in an academic text, for example. Simply

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viewing this type of text on a computer screen is not of much interest unless a qualified interpreter is present to assist the student in the library, at home or remotely. Likewise, videos with automatic captions can facilitate communication. However, in reality, the captioning may be out of synchronization with the topic being discussed in the video. It is essential that tutors encourage feedback on the effectiveness of the material made available online. A deaf or HH person who faces countless obstacles in their personal progress may feel fundamentally discouraged. Deaf and HH learners may also prefer learning in interactive groups and having note-taking interpretation and assistance because they are able to use visual clues by observing the faces and expressions of other learners and tutors. Such holistic learning can be an integral part of the process of building meaning and understanding beyond words. Of course, online learning does not exclude the possibility of group meetings, face to face. It is also very important to avoid stereotypes about deaf and HH learners. Indeed, deaf learners are far from constituting a homogeneous group whose members all have identical needs. On the contrary, any set of support measures made available to them must be tailored to individual needs and preferences.

Deaf learning concerns the process of teaching deaf people with deafness or hearing loss. The latter does not only require adapted educational techniques on the individual and group level, but also needs a specific material and environment allowing the deaf learners to independently assimilate the educational content to allow them to have the same learning level of non-deaf students.

For this, education systems for the deaf should involve IoT and AR technologies to achieve the needs and services necessary for deaf students and their educators without having to resort to voice interpreters, visual cues or note takers.

In addition to the functionalities offered by the IoT and AR equipment, the latter must also positively influence the degree of engagement of deaf students in the lectures by minimizing the auditory and visual distraction of the deaf students and also their agitation or movement in the room.

The needs of deaf students mainly concern their levels of engagement in school activities. This includes their participation in the discussions, their ability to follow the teacher's instructions, and their degree of concentration and accomplishment of school tasks.

The Internet has become an essential communication tool for deaf people. It has far greater importance than the telephone for hearing people. The webcam promotes an authentic exchange, linked to emotions. The expression of emotions is visually received by the interlocutor in the same way as the intonation of the voice informs about feelings. The use of webcams allows a rich exchange, since a real dialogue takes place through an interposed camera. The number of connected objects is growing, and its uses are multiplying, including smartphones, tablets, connected watches and bracelets, and tactile and interactive tables [3]. These connected objects are the main players in the Internet of Things (IoT). The development of the IoT tends to tie technology more deeply with the environment. Through everyday objects, technology supports individuals at every moment and in every action, especially regarding the learning of deaf persons.

Indeed, the IoT can be an excellent medium for the social integration of deaf people. Easy access to information and communication are the two most important

reasons for their use of the Internet. For them, the Internet represents accessibility, autonomy and personal development. This explains the appropriation of the Internet by deaf persons as a tool to support communication, training and empowerment. Our study leads us to conclude that the "deaf" uses of the IoT—in terms of access to information related to deaf identity, communication and training—allow deaf people to assert their identity as a group for better integration into society.

It is first and foremost by promoting relations with the hearing world that the Internet and the IoT play a role in the social integration of deaf people. Indeed, the Internet and the IoT help deaf persons to understand the hearing world. As mentioned earlier, the use of the Internet offers deaf people fertile ground for communication with hearing people. If we believe that social integration involves being able to understand and be understood and that the handicap of deafness is linked to a mode of communication different from the majority, we can conclude that Internet mediation promotes deaf users' social integration by allowing easier access to the hearing world and a better understanding of the hearing society.

Additionally, another technology seems to be very promising when used for deaf learning is augmented reality (AR). AR is the superimposition of reality and other media elements (sounds, 2D and 3D images, videos, etc.) [4, 5] calculated by a computer system in real time. It often refers to the different methods that allow the realistic embedding of virtual objects in a sequence of images. It also applies to visual perception (the superimposition of virtual images to real ones) and to proprioceptive perceptions, such as tactile or auditory perceptions. These applications are multiple and affect more and more numerous fields, such as video games, education, cinema, industries (conception, design, maintenance, assembly, piloting, robotics, etc.) and medical fields. AR is one of the emerging phenomena enabled by the development and democratization of information and communication technologies (ICT). It is involved in certain augmented forms of collaborative work and the collaborative economy. AR inserts synthetic images into real-world images using a cell phone camera or specific video glasses. Small cameras located in the middle and outside each lens send continuous video images to two liquid crystal displays on the inside of the glasses via a mobile processor. When connected to a smartphone or computer, the glasses combine computer data with live filmed reality, creating a unique stereoscopic field of view on the LCD screen, where synthetic images overlap with those of the real world. The use of AR in deaf learning will be discussed later in the paper.

A major question that our study tries to answer is: can IoT be used with AR to solve learning challenges?

Indeed, AR is an emerging technique, qualified as the technology allowing to better use the potential of the IoT. AR takes advantage of data issued from the IoT infrastructure to facilitate the visualization of the real context. Therefore, IoT-AR deployment strategies must be developed by companies to take advantage of both technologies. Moreover, IoT facilitates connecting virtual and physical contexts. Indeed, data are gathered from IoT objects, transformed into information and instantly integrated into the AR application to augment the real context.

Another major question of this study is the interest of AR-IoT models in context of learning. In this regard, due to rapid social changes, education is a field that is rapidly

evolving. IoT with AR enables revolutionary progress in education processes aligned with the new demands of society, teachers and learners.

Therefore, the IoT uses the physical objects in the educational environment to establish a more efficient learning process. This allows an effective synergy between the digital and physical environment, which facilitates the understanding of educational situations by students and facilitates group learning. Using IoT, students can collect information about homework completion time, feelings, and degree of satisfaction of students. This helps teachers to measure their methodologies and personalize their interventions to solve students' difficulties.

Another educational issue can be easily solved by IoT-AR synergy. It concerns the visualization of pedagogical learning situations that become realizable and rapidly understandable using AR which brings life to IoT data. On the other side, the IoT feeds the AR augmented scenes in real time. This creates a visual and intelligent learning environment that allows an interactive learner participation.

The research questions considered in this study in relation to the use of the IoT and AR in deaf learning are as follows:

- What are the appropriation modes for promoting deaf people's communications and social integration?
- How do deaf people use the IoT and AR in practice?
- What role does deaf identity play in the social construction of the uses of the IoT and AR?
- What place does the use of the IoT and AR take in the daily communication practices of deaf users, and what are its consequences on their integration into society?

The objectives of the research are as follows:

- Draw up a portrait of the uses of the Internet and IoT devices by deaf people to understand their contribution to deaf learning.
- Analyze how the daily context of the use of the IoT and AR affects the construction and nature of the social connections of deaf persons.
- Understand how these uses transform the way deaf people communicate on a daily basis and, more generally, the way they integrate into society.

The rest of the paper contains the following sections: Section. 2 presents a literature review. Section 3 illustrates the IoT methodology used for deaf learning. Section 4 presents the numerical results, and Section 5 provides a discussion. Then, Section. 3.3 describes the suggested deaf intelligent learning model. Finally, Section 6 concludes the paper.

## 2 Literature review

Deafness represents the incapacity of an individual to hear and recognize voices. The vast majority of HH persons are additionally discourse-disabled, and they utilize gesture-based communication (sign language) as the primary method of interacting

with others [5]. Communication through signing utilizes hand gestures, looks, lip examples and non-verbal communication rather than sound ones. Like communication with dialects, gesture-based communications differ among nations, which have their own jargon and sentence structures [6].

An advanced education for HH people is a main aspect in every country's schooling improvement. HH understudies in advanced education are a source of information about the boundaries these learners face in getting content from lectures. A deep exploration should be achieved about the adequacy of help administrations—such as deciphering, notes composing, examining and understanding courses—which affect their scholarly accomplishment [7]. E-learning is a valuable method used for educational reasons. It incorporates schooling for HH learners.

One of the main significant effects of technological advancement has been achieved in schooling and learning domains by the invention of e-learning, which is extremely advantageous for individuals in various demographical areas who cannot proceed with formal classic education [8]. E-learning frameworks help HH individuals to communicate with hearing individuals. However, the majority of existing e-learning platforms are not easily accessible and helpful for HH learners in light of their components and content, which are not adequate for HH learners. These learners frequently experience issues in getting access to the data and using it appropriately [9].

Actually, e-learning provides new possibilities for ICT in training by utilizing numerous multimedia types, such as images, video and sounds. E-learning can be achieved inside or outside lecture halls [10]. All categories of persons in the community may benefit from the e-learning advantages, including incapacitated individuals [11].

Indeed, specialists have utilized new technologies like AR to help HH people in various fields, such as medical care and education. In [12] introduced a virtual reality (VR) homeroom called "DAVEE." It helps HH people to pose inquiries and receive answers in each virtual class. Likewise, they can communicate with other learners in this virtual class and record it for asynchronous learning. The outcome of the AR experience showed the impacts of utilizing AR to develop HH learners' correspondences in the class. In [13] fostered a framework for deciphering discourse, changing it to text and showing it as captions in an AR environment. This framework helps HH people go to live theaters and comprehend the exchanges. In light of their outcomes, more HH people want to take care of and appreciate live sessions by utilizing their framework. In [14] utilized AR to further develop enlistment abilities among HH children. They showed that AR can help HH children to better master school abilities. Additionally, this platform is useful for HH people's future relations and communications with society. Researchers have shown that haptic and wearable gadgets are exceptionally valuable for HH people, particularly in the field of sound awareness.

Researchers in [15, 16, 17, 18 and 19] showed that we can utilize clever ear-worn gadgets to move material data on the two ears. Likewise, in [18] and [19] planned and assessed a head-mounted material model and a vibro-tactile AR device to help situational and spatial awareness among persons. However, none of these specialists tried their gadgets on deaf or HH people. In [20] assessed

representations for spatially finding sound on AR devices and giving consciousness of natural sound on visual presentations to HH people. They showed that people with hearing issues utilize obvious signs to perceive and recognize sounds. In [21], an online review is led to examine inclination for versatile and wearable sound mindfulness frameworks among HH members. They showed that practically all members needed both visual and haptic input gadgets. In [22], authors planned a cell phone-based acoustic detecting and warning framework called "UbiEar" and showed that this framework can help HH learners to familiarize themselves with significant acoustic events in their everyday lives.

Some other analysts utilized a haptic input framework to instigate the sensation of tap moving to HH people [23]. They showed that HH people who utilized their framework had a totally different sensation of tap moving and delighted in it, regardless of whether they could hear the sound. In [24] created a framework for HH people called "MuSS-Bits," which helps them feel instruments of sounds on their skin utilizing haptic criticism. Another research designed an exceptional gauntlet for HH people that can decipher gesture-based communication to discourse. It further develops HH people's interchanges and collaborations in AR conditions [25]. There are numerous cases, for example, LIVEJACKET [26], that showed the impacts of utilizing haptic criticism gadgets for HH people.

The study in [27] provides an educational environment for teaching the IoT security. The study presents a set of data in the form of an IoT security database. A real prototyping is achieved on an online educational system based on AR. However, during the prototyping, certain assumptions were not considered, such as taking into account the functional components of the used system. Moreover, there is no prototyping results.

The study in [28] focuses on the contribution of the IoT and AR in the construction of intelligent learning involving a set of complex communications using devices such as smartphones, watches and glasses. However, the investigated studies were not criticized and discussed.

In [29], a smart augmented reality concept is introduced in mobile phones to enhance the learning process in which students follow augmented 3D models established by the teacher. It is shown by experiments that the subjects are explained by 3D AR models better than classical teaching methods. Moreover, the proposed system is easily deployable in android devices and supports the recognition of multiple targets. On the other hand, this AR system must be combined by AI to obtain smarter automated applications.

Authors in [30] introduces an intelligent learning system based on game theory technique. This game model uses and integrates mobile IoT, AR and web techniques in the context of e-learning. By interacting in the intelligent classroom, this system determines the interests, impressions and degree of motivation of the learners. To assess the knowledge of students, this system offers an IoT course on Moodle in the University of Belgrade. Unfortunately, the size of the sample of evaluation is limited. Moreover, this learning system does not offer a module that personalizes the learning process and does not integrate student participations in the configuration and development of the courses.

The study in [31] suggests a mobile AR system to learn telugu language. This system uses the smartphone's camera to identify external objects and then augments the real images with virtual information to facilitate the language learning. A set of sampling tests were achieved to assess the efficiency of the system in the learning of language. On the other hand, the efficiency of the proposed system must be verified with higher number of users.

Authors in [32] suggest an IoT-AR-based technologies prototyping to enhance the learning process of student having learning problems. Unfortunately, this prototype is not experimentally tested on a real classroom. Moreover, a questionnaire on a set of undergraduate students can be achieved as a quantitative analysis.

Another study in [33] proposes the integration of AR and IoT for the learning of students. A set of 90 learners divided in two groups were used in experiments of 120 min. The findings of the study illustrate the improvement of the knowledge about the air pollution. Moreover, by comparing the degree of satisfaction of two groups of students; a first group using VR and a second one using the IoT; it was noticed that there is no difference in the level of satisfaction of the two groups.

Recently, utilizing various sorts of wearable gadgets in AR has become exceptionally normal. Unfortunately, most of these gadgets are intended for people with no hearing issues. Moreover, there is no thorough examination of the convenience of these gadgets for HH people. The significant point in the majority of these cases is that HH people can utilize them; however, they cannot appreciate them like people without hearing issues.

AR and IoT learning Media is a relevant component which is necessary to enhance of the learning quality. In this context, numerous studies suggest to design and use AR learning media to address the students learning issues:

In [34], a AR learning media based on elementary mathematics is used as a learning medium to learn geometry in elementary schools. This study deduces that AR learning media helps teachers to increase the motivation and enthusiasm of students in the learning process. Moreover, the AR media allows the learning outside the school.

The study in [35] suggests an android platform for geometry learning using AR media. The proposed system was designed to teach three-dimensional objects for high school students. The aims of the study were to evaluate the efficiency of the way the student uses the AR media, and to the degree of comprehension of the geometry concepts. The findings of this study illustrates that the use of the proposed android-based AR learning media enhances the positive responses of the students and the learning of the three-dimensional geometrical concepts.

Another study in [36] introduces a development of an AR learning media for the learning of electricity related concepts. The study suggests the preliminarily results using a single pretest–posttest group of thirty learners from the first-year classes in a high school in Garut. The experimental process reveals that the proposed AR learning media improves the understanding of students.

In [37], the authors introduce a design of a smartphone and AR-based learning media to increase the autonomy of the students during the learning process. The used mobile media involves android smartphones, tablets and cameras features to run AR technology. To measure the efficiency of the used media, a set of

questionnaires were proposed to learners and teachers. The results shows that the use of AR learning media rises the user's learning autonomy.

In most of these studies, a unique AR context or gadget should be intended for HH people. AR engineers should utilize captions, images, interpreters, features, and so forth in the AR environment to make it usable for HH people. In actual systems, HH people cannot utilize AR applications that are intended for people without hearing issues. Contrary to other studies investigated in this section, our study is based on experimental real data issued from a case of study and a survey. It also suggests a hybrid IoT and AR intelligent E-learning environment.

## 3 The internet of things and augmented reality methodology for deaf learning

### 3.1 The internet of things for deaf persons

The IoT represents a "network of networks" of connected objects, identifiable in a unified way so that we can act on them or know their state [38, 39]. It designates various technical solutions (RFID, TCP/IP, mobile technologies, etc.) whose role is to identify objects, capture, store, process and transfer data in physical environments and between physical and virtual environments. The major challenge is not to invent new technologies but simply to adapt already existing ones in order to connect and integrate them. Connected objects are equipped with sensors to measure environmental parameters and exchange the data generated between objects. As illustrated in Fig. 1, this data ultimately travels to service platforms for storage and analysis. Figure 2 illustrates the proposed AIoT architecture.

The IoT can be easily used in different types of applications. For example, sensors in lamps can be used to make buildings smarter. Moreover, the IoT allows easier control of the mobility of objects. Through the use of control chips, actuators and motors are integrated into the same design. As these IoT components are integrated into a common environment, there are some challenges regarding communication interfaces, noise levels and power consumption. We must therefore investigate the possible articulations of the IoT and connected objects to better support

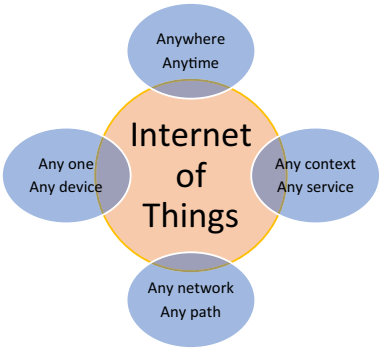



**Fig. 1** Illustration of the IoT concepts

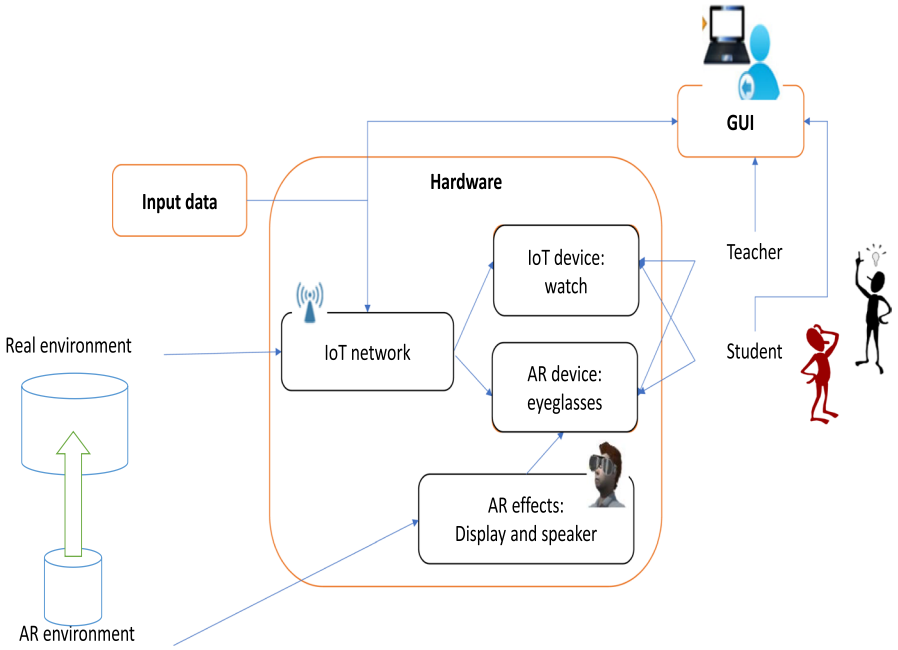



**Fig. 2** Proposed AIoT architecture

learning through the IoT. There are three dimensions of analysis: data, interfaces and pervasiveness.

**Data**: The functions of connected objects are based on data corresponding to environmental parameters collected through sensors.

**Interfaces**: Human–machine interfaces tend to bring together the methods of controlling and entering information with its representation: interfaces define how humans interact with connected objects.

**Pervasiveness**: The growing entanglement of digital and physical environments tends to give ICT a ubiquitous character: objects exchange with each other and with humans having access to information and digital tools without time and space constraints.

### 3.2 Augmented reality and the internet of things for deaf persons

The hybrid IoT-AR model can be justified as follows: The advantages of using IoT in e-learning can be summarized as follows: it automates most of the teaching tasks and subsequently it reduces costs, it allows the improvement of learners' productivity. Besides, the IoT optimizes the use of educational resources and anticipates the needs of students by obtaining information from the educational context. This information allows the rapid identification of problems of students.

On the other hand, the use of AR in e-learning has several advantages which can be summarized as follows: AR promotes communication and student engagement

for a more optimized experience. Moreover, in learning as in business, AR increases the responsiveness, visualization, and implementation of services by increasing the knowledge of users that should be often personalized. This enhances the user experience and subsequently makes learning easier and more comfortable.

AR is an innovation with huge promise. It is arising as the key that will open the maximum capacity of the IoT. AR applications are utilizing the numerous types of information created by IoT gadgets and segments to assist IoT workers to be more efficient.

Indeed, AR offers a helpful and natural way for clients to imagine and collaborate with IoT objects and their related information. Besides, AR services are made conceivable by utilizing and taking advantage of the more refined data context offered by IoT devices.

AR and the IoT have received huge consideration as a key to empowering innovations for making educational environments more intelligent and intuitive. AR is a kind of intuitive medium that gives a perspective to reality in an increased space by spatially enlisting valuable produced data to assist individuals in understanding the world and intensifying their insight for tackling issues. The IoT represents a set of connected gadgets and devices installed with minimum processing components for detecting and gathering data and communicating.

### 3.3 Proposed intelligent learning model

The intelligent learning model that we propose in this study is based on the IoT by treating the three dimensions of analysis (data, interfaces and pervasiveness). The interweaving of the digital with the physical environment operated by the IoT also promotes the implementation of authentic situations where the object of learning is contextualized. Learning, in which knowledge is constructed, is important on the real and social levels. On the other hand, the use of IoT objects for educational objectives has been theorized in several studies agreeing on three points: the need to learn by experience, the need to be active, and the model of interactions between learning objects and people [40].

The IoT data help to better understand the learner. Connected objects and their tangible interfaces interact with the environment and provide the ability to control learners. The pervasiveness of the IoT reduces the constraints of time, space and support.

The reduction in component costs and their miniaturization combined with the concreteness of the data allow the use of connected objects to understand the learning environment. Using the sensors of connected objects (GSM, tablet, etc.) and self-assembly component kits (Arduino, Raspberry, etc.), learners have accessible measurement tools to contextualize the learning.

The data generated by connected objects provides information about the activities and movements of the learner, allowing monitoring beyond digital training devices. This information is related to activities of informal learning: reading a book on an online library using the IoT, watching a documentary or attending a conference. The

learning and training systems can refine their models and improve the monitoring process.

The design process of each phase in the proposed AIoT-based learning system is represented by Fig. 3.

To overcome the challenges that deaf students face (as discussed above), the deaf intelligent learning model is proposed. The aim of this model is to enrich deaf learners with meaningful learning that is extracted from learning objects and achieve effective communication between deaf learners and teachers. Meaningful learning is described as a means to increase students' knowledge regarding learning objects by using visual and auditory resources [41]. The suggested model, which is based on the IoT network, consists of the following phases, as shown in Fig. 4:

- Phase 1: Speech recognition

The speech acquired from the teacher is recognized and converted to text using a speech recognition system to facilitate the next phase. The suggested speech recognition system relies on an intelligent algorithm that recognizes spoken vocabulary items. This algorithm uses the speech audio signal to identify the acoustic indices. Then, it computes the probability of the index string being caused by a particular hypothesized utterance element.

- Phase 2: Text summarization

One of the aforementioned challenges is the difficulty of understanding long texts due to the low working memory limit of deaf students. This problem can be solved

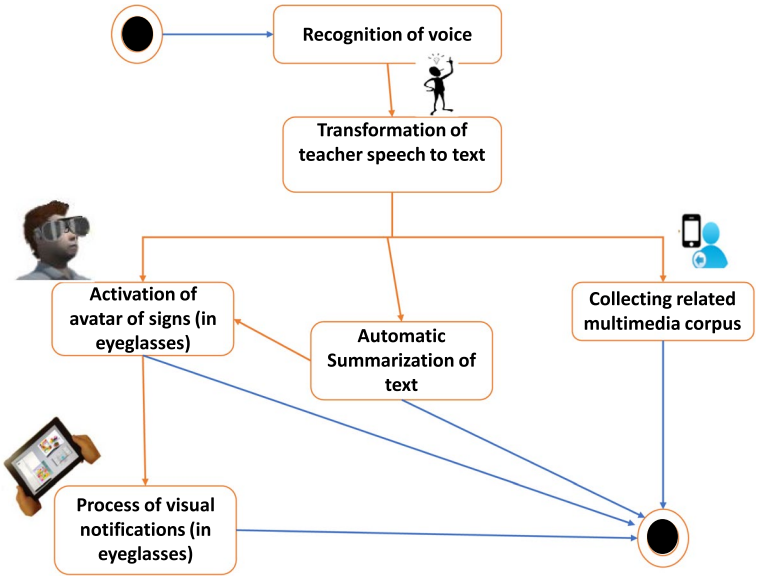



**Fig. 3** Functionalities proposed in the suggested AIoT (AR + IoT) learning system

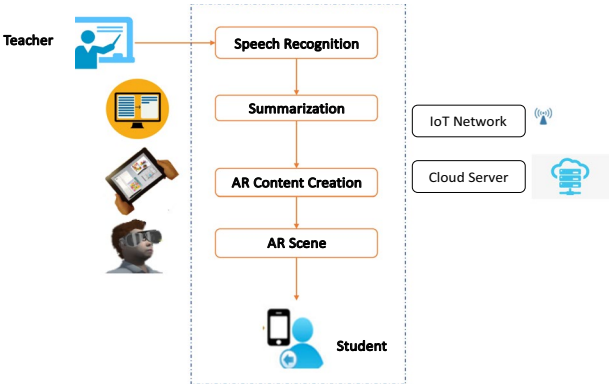



**Fig. 4** Deaf intelligent learning model based on augmented reality and the internet of things

by applying the technique of text summarization, which helps in producing a summary of the input text that contains the main idea. This phase relies on a process that collects the relevant information from the text to construct a reduced version of the text. In this process, the words are not reformulated; they are selected and rearranged to build a new shorter structure of the text.

- Phase 3: Augmented reality content creation

The AR content creation phase is responsible for creating virtual elements that are related to the summarized text from the cloud server. This phase relies on collecting multimedia elements from different sources (from the cloud) in relation to the summarized text. The text is analyzed, and relevant keywords are extracted to be used in the search.

- Phase 4: Augmented reality scene

The AR scene phase is responsible for visualizing the created VR elements on the wearable VR glasses. This phase is based on the use of AR hardware, which transforms the VR elements into an AR scene. Image classification algorithms can be incorporated into the phases of the suggested deaf intelligent learning model to enhance its accuracy and efficiency.

Indeed, connected objects are accompanied by new human–machine interfaces that can be used in learning to control activities promoting bodily engagement and interaction with the environment. Among the human–machine interfaces of connected objects, we can distinguish tangible user interfaces (TUI) and surface user interfaces (SUI). With TUIs, information takes the form of a tangible object that the user can manipulate. A robot for learning programming without code and without screen is an example of a TUI. By ordering parts corresponding to instructions (forward, backward, etc.), it is possible to program and direct the robot. SUIs refer to the interfaces of interactive tablets and whiteboards: it is a screen or a projection,

horizontal or vertical [39]. The use of sensors, virtual or AR, and ambient interfaces add an interactive layer to the environment and improve the learning process. LEDs indicating the ON / OFF position of electronic devices and the use of radio frequency identification (RFID) and near field communication (NFC) tags are examples of IoT objects [42]. Due to the pervasiveness of connected objects, learning and control activities are part of mobility. The learning and control experience persists through the supports used by the learner, and the context is taken into account by training systems. Improving the performance of connected objects (computing power, bandwidth, battery, etc.) makes training devices and educational resources accessible and usable on mobile devices, since mobility is becoming an aspect of online learning.

These conditions encourage new practices, such as micro-learning and "Bring Your Own Device." Micro-learning refers to the distribution of a course on a mobile device in the form of several short units. For education, it is good practice to invite students to use their personal devices rather than those of the institution [43].

## 4 Results

### 4.1 Data collection

We envisioned the data collection in two stages. The first step involved the technique of logbooks that our participants had to fill out. The information collected in these logbooks was intended to provide an initial portrait of the Internet use by deaf people and to better define the questions proposed to the participants. The second stage of data collection involved semi-structured interviews with the same participants.

#### 4.1.1 Survey

The questionnaire method seems to be more judicious in obtaining information about the use of the Internet by deaf persons. Considering that culturally deaf people are very direct, more precise answer choices involving fewer personal data seem more appropriate. The questionnaire (Appendix I) asks participants for information about their type of use and frequencies of use of IoT and AR devices. We decided not to ask for precise information about the visited sites, but rather we distinguished between different types of use that we presented in several categories. These types of use concern the search for information, viewing of videos, frequentation of social networks, communication, entertainment, training, viewing news sites, viewing deafness sites, and viewing utility and commerce sites. For each of these categories, participants were asked to rate their frequency of use (daily, several times a week, once a week, rarely or never) and their average durations of use. We believe that this tool was more culturally appropriate for our participants than the logbook, for which we only had one feedback. Of course, several participants did not complete the questionnaire. We followed up twice with the participants but did not want to insist, as their participation in this research was voluntary, and they were not forced to respond. It should also be noted that we received the questionnaires after conducting

the interviews, whereas we would have liked to receive them before to be able to better identify the user profiles prior to the interviews. At the end, the questionnaires received provided us with valuable information that seems consistent with what was said in the interviews.

#### 4.1.2 Selection of respondents

The participants in our research were recruited using the so-called snowball method [44]. This method consists of identifying a first person who meets our selection criteria and who agrees to participate in our research. This person is asked to recommend another close person who also meets these selection criteria and who is likely to contribute to our research. Ideally, this second participant will in turn recommend someone close to them and so on. One advantage of this recruitment technique is that it speeds up the selection process, which could otherwise be quite lengthy. Three criteria for selecting participants guided the recruitment. Firstly, the participants had to have a sufficient level of literacy to be able to use the Internet in a general way and to express themselves in writing. The second criterion implied a certain mastery of the daily use of the Internet. Finally, all participants had to be over 18 for ethical reasons.

### 4.2 Data analysis

The objective of the analysis of the completed questionnaires was to identify the user "patterns" based on the frequencies and types of use of the participants. This descriptive information supplemented the information gathered during the interviews. The analysis of the interviews aimed to grasp the meanings of uses related to these patterns and to identify the place of the Internet in the daily communication practices of the participants.

The first step of analysis was the transcription of the data. This step allows us to appropriate the content in order to codify and analyze it. The transcription stage allowed us to do a pre-analysis, which revealed the recurring response elements both within the same interview and between interviews. Following the transcription of the interviews, we proceeded to the second step, which consisted of an in-depth read of the content, line by line. At the same time, we created summary sheets specific to each interview. The summary sheet allowed the researcher to summarize the addressed themes and identify the important elements and particularities of the participants.

The third step consisted of coding the transcripts to determine the main categories of learning in the collected data and highlight those that were recurrent. The coding was conducted by analysis aid software, and then we analyzed each interview first individually, then using a cross-sectional analysis of all the interviews to identify the similarities and disparities among the participants.

The results of the application of the survey to 126 participants are shown in Table 1 and the figures below.

**Table 1** Age of participants

| Age interval | 18–25 | 25–30 | 30–35 | 35–40 | +40 |
|---|---|---|---|---|---|
| Percentage of participants | 44,38% | 22,46% | 21,63% | 6,23% | 5,29% |

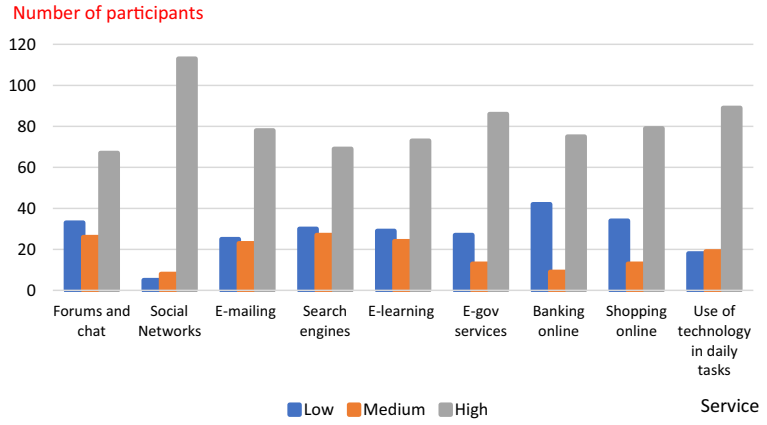



**Fig. 5** Degree of use of the services

The results (Figs. 5, 6, 7, 8, 9, 10) showed that the attentional capacities of deaf people can undoubtedly be redirected by visual periphery distractors due to their hearing problems. Online modules through a reasonable course with interactive content could help deaf persons in the learning process. Besides, most deaf people give signs of a low working memory limit due to motion-based communications. Thus, learning modules should not be incredibly long, and an effort should be made to isolate the learning content into short modules, which can be more easily followed by the learners. Deaf people seem to imagine that understanding texts is a challenging task due to partitions in their working memory and the absence of hear-capable information. The latter is the central driver of their veritable difficulties in text creation. Therefore, the presentation of the learning modules is a fundamental step for improving the educational outcomes for deaf persons.

The results have shown that course information and content should be presented in a visual way for nearly deaf and hearing weakened learners using pictures, chronicles with subtitles and, clearly, correspondence by means of signal accounts, as they can deal with pictures and chronicles more significantly than words. Hence, it is apparent that the informative material should similarly be given in correspondence through motions chronicles, as this is the best strategy to help deaf persons in the learning process.

Regarding evaluation strategies, HH persons had positive viewpoints toward the presence of exercises after an online course to complete their adjusting needs. Given this, an e-learning system should provide an electronic examination system, considering certain completion principles for self-appraisal.

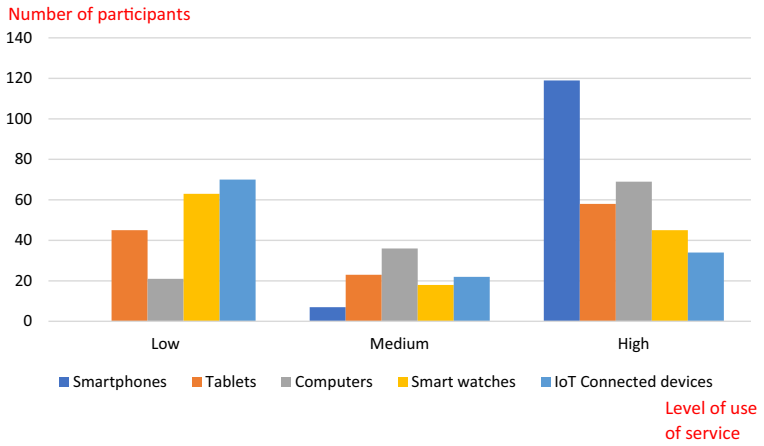



**Fig. 6** Degree of use of intelligent devices

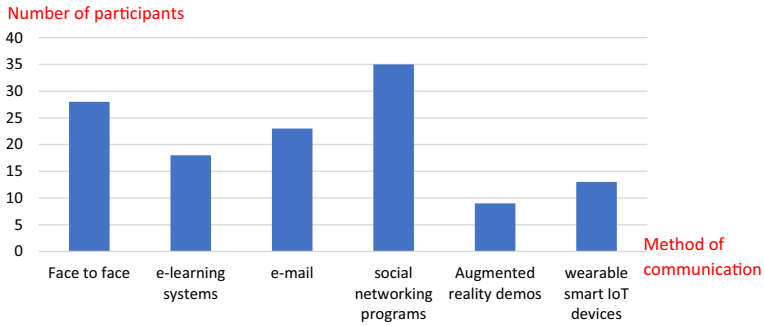



**Fig. 7** Communication with teachers in presential courses

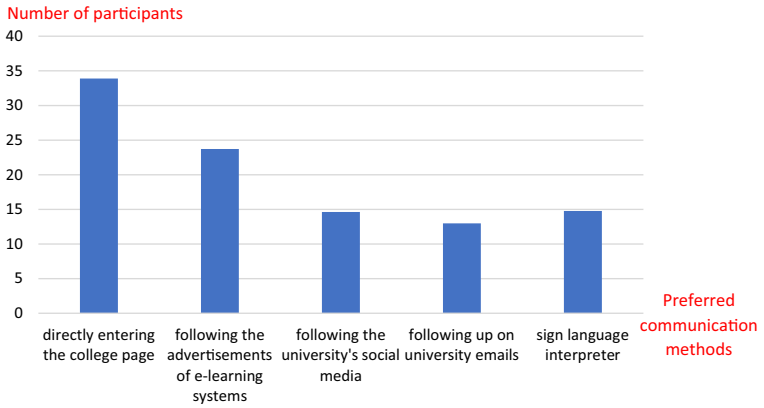



**Fig. 8** Preferred method of communication to follow up on university activities and announcements

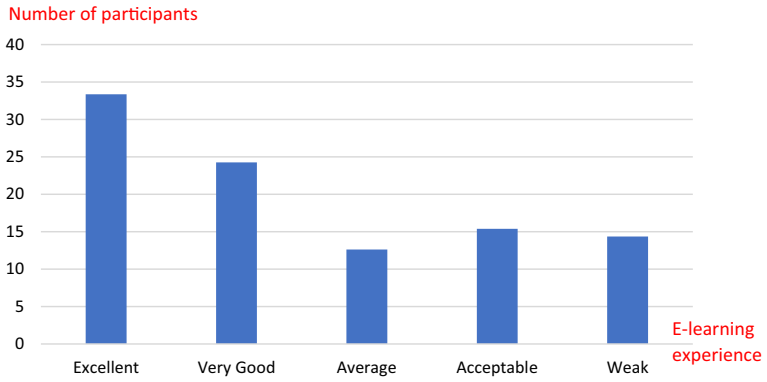



**Fig. 9** Assessment of the E-learning experience of participants

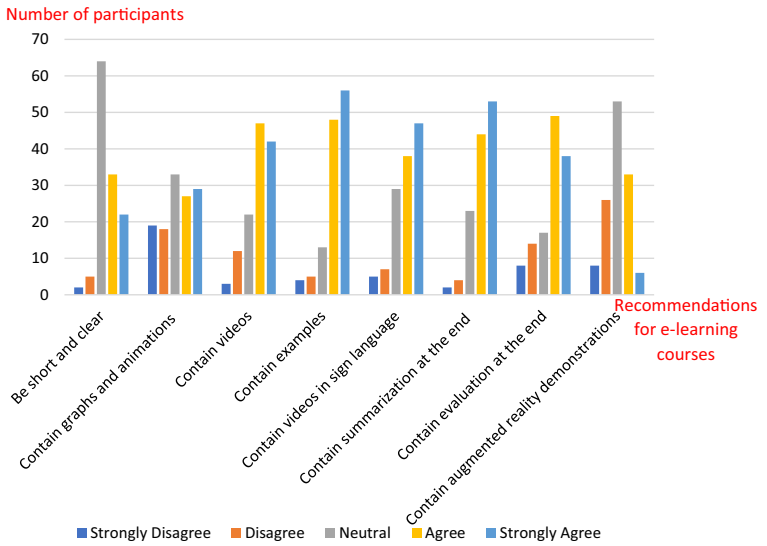



**Fig. 10** Recommendations regarding the attended e-learning courses

As demonstrated by the results, individuals imparted inspiring viewpoints toward the intelligibility of online modules. Considering this reality, the use of long messages should be avoided since deaf persons have low working memory limits.

### 4.3 Results of speech summarization process

To summarize the speech, we use cascaded learning [45] which is achieved by a cascade of text summarization and speech recognition. We compare this model (Speech2Concept) with the Text2Concept model [46] in which semantic concept learning is used to perform multimodal speech synthesis. Speech is then represented by concepts, afterward by a synthesis model that generates summaries. The

**Table 2** Evaluation des modèles d'apprentissage

| Model | F-1 | Recall | Precision |
|---|---|---|---|
| Speech2Concept | 58.8 | 55.8 | 62.3 |
| Text2Concept [46] | 54.8 | 57.3 | 52.5 |

**Table 3** Evaluation of the responsiveness of the students

| | Normal set of deaf students | Deaf students using IoT devices | Deaf students using AR | Deaf students using IoT and AR |
|---|---|---|---|---|
| Average time (seconds) of response per question on exams | 73 | 58 | 64 | 41 |
| Average grades (per 100) | 72.63 | 84.93 | 81.97 | 88.21 |
| Percentage of misunderstood questions | 45.32% | 29.91% | 32.49% | 12.86% |

evaluation of learning models is done with F1 measures. Recall and Precision in Table 2.

According to the results of Table 2, the proposed module gives better results than the Text2Concept module.

### 4.4 Case of study

Table 3 compares the responsiveness of a set of deaf students. Four sets of students are proposed: Standard learners (normal set of students) without use of AR and IoT technologies; Students using only IoT technologies; Students using only AR technologies; Students using both IoT and AR technologies.

Table 3 clearly shows the contribution of IoT and AR technologies on enhancing the time of response and the grades of the students.

### 4.5 Statistical hypothesis testing from questionnaire

To study the interaction of the students with the proposed IoT-AR learning model and the influence of the latter on the educational process, a statistical hypothesis testing is used based on the performed survey.

**H1** Deaf students believe that their degree of understanding is increased when using IoT-AR system.

>>H1 measures the degree of improvement of understanding of students by using IoT-AR system.

**H2** Deaf students believe that the IoT-AR system are difficult to use.

>>H2 measures the degree de difficulty of use.

**Table 4** Standard deviation and mean score values of the questionnaire statements

| | Standard deviation | Mean score |
|---|---|---|
| DU: degree of understanding (H1) | 0.64 | 3.47 |
| DF: degree of difficulty of use (H2) | 0.42 | 3.85 |
| DP: degree of pleasure of use (H3) | 0.67 | 4.26 |
| DS: degree of satisfaction (H4) | 0.73 | 2.92 |
| DA: degree of acceptance (H5) | 0.89 | 4.94 |

**Table 5** Study of relationship between pairs of variables using regression

| Variable 1 | Variable 2 | $p$ value | $R^2$ | Coefficient |
|---|---|---|---|---|
| DU | DS | $<0.05$ | | 0.36 |
| DF | DS | $<0.04$ | | 0.58 |
| DP | DS | $>0.06$ | | 0.23 |

**H3** Deaf students believe that the IoT-AR systems are pleasant to use.
>>H3 measures the degree of Interest and how much the IoT-AR system is delightful.

**H4** Deaf students believe that the proposed IoT-AR system is satisfying.
>>H4 measures the overall satisfaction degree.

**H5** Deaf students accept the proposed IoT-AR system.
>>H5 measures the intention to use the system.

To examine the statements of the questionnaire, the standard deviation and mean score values are measured as follows in Table 4. Table 4 gives a descriptive statistic assessment of the identified hypothesis.

### 4.6 Regression analysis

In this section, we use regression to test the first two hypotheses H1 and H2 and study the relationship between pairs of the model. The value of α is set to 0.05. Table 5 summarizes the regression results.

Since the $p$-values are lower than α for the dependence DU-DS and DF-DS, then the dependence is confirmed between the degree of understanding and degree of satisfaction; and between difficulty of use and degree of satisfaction.

However, since the p-values are higher than α for the dependence DP-DS, then there is no dependence between of pleasure of use and degree of satisfaction.

### 4.7 Testing null-hypothesis

Let's $z^2 = \sum_{i-=1}^{n} (OBS_i - EXP_i)^2/E_i$ be a hypothesis that evaluates the behavior of the deaf students. It assesses the difference between observed values *OBSi* and expected *EXPi* ones. The following Null hypotheses are considered:

**NH1** The degree of use of online services does not affect the relevance of IoT-AR system for deaf students.

**NH2** The degree of use of devices does not affect the relevance of IoT-AR system for deaf students.

**NH3** The method of communication of deaf learners with teachers does not affect the relevance of IoT-AR system for deaf students.

As an example, the result of evaluation of NH2 are described in Table 6.

The null-hypothesis NH2 is rejected with a considerable significance level. Hence, it can be confirmed that the frequency of use of smart devices by deaf students has an influence on the importance of a possible IoT-AR system for learning.

## 5 Discussion

The current study concerns an exploratory examination of deaf individuals regarding their interactions with e-learning systems. The study objectives were to grasp the deaf scholarly characteristics and points of view in order to identify the way in which they learn better and implement more appropriate e-learning systems for deaf persons.

Regarding ICT techniques, the assessment of deaf individuals seems to support PDA, tablets and new devices more than PCs. For the web, for example, deaf people communicate through interfaces, electronic media, email and getting news. They participate in e-learning, e-banking and entertainment activities less frequently.

These results demonstrate that there are no apparent differences among participants regarding their gender or cultural background. Moreover, the results demonstrate that there are no differences between inborn deaf individuals and people who lost their hearing after birth. Nevertheless, age, social and educational status

**Table 6** Analysis of results of question 8 for the evaluation of NH2

| | Smartphones (%) | Tablets (%) | Computers (%) | Smart watches (%) | IoT Connected devices (%) |
|---|---|---|---|---|---|
| Low | 5 | 9 | 2 | 14 | 34 |
| Medium | 12 | 13 | 14 | 21 | 31 |
| High | 83 | 78 | 84 | 65 | 35 |

appeared to essentially influence the e-learning motivations of deaf persons. Furthermore, the use of new IoT and AR-based technologies and devices seems to be very attractive for deaf persons and contributes to enhancing the learning process for them. Moreover, these technologies seem to ameliorate the way deaf persons interact and communicate with society.

Practical implications of learning media are as follows: IoT and AR offer benefits individually, collectively, virtually and physically. These technologies encourage student motivation, interaction, and engagement to facilitate the achievement of their goals. Pedagogically, the design of the AR approach of learning influences the educational process which encompasses different but balanced facets of teaching techniques. On the other hand, the technical progress of the devices used in augmenting the real learning context allows the engagement of the student in the educational process. Likewise, teaching and learning practices influence the experience of learners.

## 6 Conclusion

This paper presents the use of new technologies such as the IoT and AR in deaf learning. The IoT refers to technological advancements in computer networking using real-world IoT objects that can be connected to communicate with each other over the Internet. The IoT and AR have greater potential to remove all barriers in deaf education. Therefore, this study suggests a model for deaf intelligent learning that may overcome the challenges in deaf education. Much remains to be done for the IoT and AR to make deaf learning even simpler and easier. We faced constraints of several kinds during the realization of this research. The most important constraint appeared at the stage of interpreting the results. Adaptability is also one of the central topics of this research. It has been one of our priorities throughout the data collection process. Of course, in order to interview deaf participants, we had to collaborate with sign language interpreters.

## Appendix I

Questionnaire about the use of modern communication and e-learning technologies by deaf persons.

Section (I): General information.

Q1: What is your age?

◯ 18–25.
◯ 25–30.
◯ 30–35.
◯ 35–40.
◯ More than 40.

Section (II): Use of modern communication technologies.

Q2: What is your degree of use of the following services?

| | Low | Medium | High |
|---|---|---|---|
| Forums and chat | | | |
| Social networks | | | |
| E-mailing | | | |
| Search engines | | | |
| E-learning | | | |
| E-gov services | | | |
| Banking online | | | |
| Shopping online | | | |
| Use of technology in daily tasks | | | |

Q3: What is your degree of use of the following devices?

| | Low | Medium | High |
|---|---|---|---|
| Smartphones | | | |
| Tablets | | | |
| Computers | | | |
| Smart watches | | | |
| IoT Connected devices | | | |

Q4: For presential courses, what is the best method to communicate with the teacher?

◯ Face to face.
◯ Through e-learning systems (e.g., Blackboard).
◯ Communicate via e-mail.
◯ Communicate through social networking programs (WhatsApp—Twitter—Facebook).
◯ Through intelligent wearable smart IoT devices (smart watches).

Section (III): E-learning experience

Q5: What is the preferred method of communication to know the activities and announcements of the department, college and university?

◯ By directly entering the college or university page.
◯ By following the advertisements of e-learning systems (example: Blackboard).
◯ By following the university's social media (WhatsApp—Twitter—Facebook).
◯ By following up on university emails.
◯ Via direct communication with a sign language interpreter.
◯ Other: ……

Q6: How do you qualify your e-learning experience (if any)

◯ Excellent.
◯ Very Good.
◯ Average.
◯ Acceptable.

◯ Weak.

Q7 In case you attended e-learning courses, you recommend that the course should

| | Strongly disagree | Disagree | Neutral | Agree | Strongly agree |
|---|---|---|---|---|---|
| Be short and clear | | | | | |
| Contain graphs and animations | | | | | |
| Contain videos | | | | | |
| Contain examples | | | | | |
| Contain videos in sign language | | | | | |
| Contain summarization at the end | | | | | |
| Contain evaluation at the end | | | | | |

## Appendix II

Summary of information of participants (see Tables 7, 8, 9, 10).

**Table 7** Gender of participants

| Gender | Male | Female |
|---|---|---|
| Percentage of participants | 57,28% | 42,71% |

**Table 8** Time of hearing loss

| Loss of hearing | Since birth | After birth |
|---|---|---|
| Percentage of participants | 65,21% | 34,78% |

**Table 9** Functional status

| Functional status | Student | Private sector | Public sector | Retired | Waiting for a job opportunity |
|---|---|---|---|---|---|
| Percentage of participants | 44,81% | 13,19% | 9,76% | 11,05% | 21,18% |

**Table 10** Educational status

| Level of education | Does not know how to write and read | Has no education but knows how to write and read | Primary school | Secondary school | University |
|---|---|---|---|---|---|
| Percentage of participants | 3.87% | 21.98% | 17.93% | 12.36% | 43,86% |

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